# ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** third planet from the Sun | Order of planets. |
| 2 | **(b)** troposphere | Closest to the ground. |
| 3 | **(c)** rivers | Not salty. |
| 4 | **(d)** the Moon | One Moon. |
| 5 | **(b)** stratosphere | Above the troposphere. |
| 6 | **(d)** salty | Contains salts. |
| 7 | **(a)** Earth | Known so far. |
| 8 | **(b)** ultraviolet rays | UV. |
| 9 | **(d)** glaciers and ice caps | Frozen. |
| 10 | **(c)** thin and solid | Outer layer. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Crust (1); Mantle and core (1). | 1+1 |
| 12 | All water on the Earth (2). | 2 |
| 13 | Troposphere (2). | 2 |
| 14 | Layer of gases around the Earth (2). | 2 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Troposphere (1); Stratosphere, mesosphere (1); Thermosphere, exosphere (1). | 1+1+1 |
| 16 | Diseases (1); Smog (1); Acid rain (1). | 1+1+1 |
| 17 | Rivers (1); Lakes (1); Groundwater (1). | 1+1+1 |
| | **OR:** Nitrogen 78 per cent (1); Oxygen 21 per cent (1); Other gases 1 per cent (1). | 1+1+1 |
| 18 | Outermost layer (1); Thin and solid (1); Contains rocks and soil (1). | 1+1+1 |
| | **OR:** More UV reaches the Earth (1); Skin cancer (1); Harm to crops (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Crust (1); Mantle (1); Outer core (1); Inner core (1); Features (1). | 1+1+1+1+1 |
| | **OR:** Lithosphere (1); Hydrosphere (1); Atmosphere (1); Biosphere (1); Links (1). | 1+1+1+1+1 |
| 20 | Troposphere (1); Stratosphere (1); Mesosphere (1); Thermosphere (1); Exosphere (1). | 1+1+1+1+1 |
| | **OR:** Ice melts (1); Sea level rises (1); Floods (1); Droughts (1); Loss of species (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** troposphere | Where clouds are. |
| 2 | **(b)** save water | Conserves. |
| 3 | **(d)** hot and thick | Middle layer. |
| 4 | **(b)** troposphere | Dense layer. |
| 5 | **(b)** all living things | Life zone. |
| 6 | **(a)** core | Innermost. |
| 7 | **(d)** nitrogen | 78 per cent. |
| 8 | **(b)** carbon dioxide | Air pollution. |
| 9 | **(b)** mixture of gases | Mostly N2 and O2. |
| 10 | **(b)** 21 per cent | One-fifth. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Zone of life on Earth (2). | 2 |
| 12 | Collecting and storing rainwater (2). | 2 |
| 13 | Nitrogen (1); Oxygen (1). | 1+1 |
| 14 | Carbon dioxide (1); Methane (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Water (1); Air (1); Suitable temperature (1). | 1+1+1 |
| 16 | Innermost (1); Very hot (1); Iron and nickel (1). | 1+1+1 |
| 17 | Weather is short-term (1); Climate is long-term (1); Weather changes daily (1). | 1+1+1 |
| | **OR:** Breathing (1); Protection from UV (1); Weather and rain (1). | 1+1+1 |
| 18 | Liquid water (1); Oxygen (1); Life (1). | 1+1+1 |
| | **OR:** Lithosphere: rocks (1); Hydrosphere: oceans (1); Atmosphere: air; biosphere: life (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Oxygen (1); Carbon dioxide (1); Ozone (1); Temperature (1); Weather (1). | 1+1+1+1+1 |
| | **OR:** Location (1); Absorbs UV (1); CFCs (1); Ozone hole (1); Protection (1). | 1+1+1+1+1 |
| 20 | Gases trap heat (1); Keeps Earth warm (1); CO2, methane (1); Excess causes global warming (1); Effects (1). | 1+1+1+1+1 |
| | **OR:** Water (1); Trees (1); Energy (1); Soil (1); 3 R's (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** reduce carbon dioxide | Photosynthesis. |
| 2 | **(b)** oxygen | Needed for respiration. |
| 3 | **(a)** greenhouse gas | Traps heat. |
| 4 | **(b)** 23.5 degrees | Gives seasons. |
| 5 | **(d)** carbon dioxide | Photosynthesis. |
| 6 | **(c)** rise in Earth's average temperature | Climate change. |
| 7 | **(d)** rotation | Spinning. |
| 8 | **(c)** 71 per cent of the Earth | Oceans. |
| 9 | **(b)** ozone layer | Chemicals. |
| 10 | **(b)** spherical | Slightly flattened. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Drinking (1); Farming (1). | 1+1 |
| 12 | Layer of ozone gas in the stratosphere (1); Absorbs UV rays (1). | 1+1 |
| 13 | Increase in average temperature of the Earth (2). | 2 |
| 14 | Water covers most of it (1); Looks blue from space (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Burning fuels (1); Deforestation (1); Industries (1). | 1+1+1 |
| 16 | Earth has air (1); Earth has water (1); Earth has life (1). | 1+1+1 |
| 17 | Venus: hot (1); Mars: cold (1); Earth: life (1). | 1+1+1 |
| | **OR:** Close taps (1); Harvest rain (1); Reuse water (1). | 1+1+1 |
| 18 | Give oxygen (1); Reduce CO2 (1); Reduce soil erosion (1). | 1+1+1 |
| | **OR:** Rises (1); Ice melts (1); Sea levels rise (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Distance (1); Water (1); Oxygen (1); Temperature (1); Magnetic field (1). | 1+1+1+1+1 |
| | **OR:** Spherical (1); Radius about 6400 km (1); Age 4.5 billion years (1); Rotation (1); Revolution (1). | 1+1+1+1+1 |
| 20 | Oceans (1); Ice (1); Rivers (1); Groundwater (1); Fresh water is little (1). | 1+1+1+1+1 |
| | **OR:** Smoke (1); Vehicles (1); Diseases (1); Acid rain (1); Control (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.